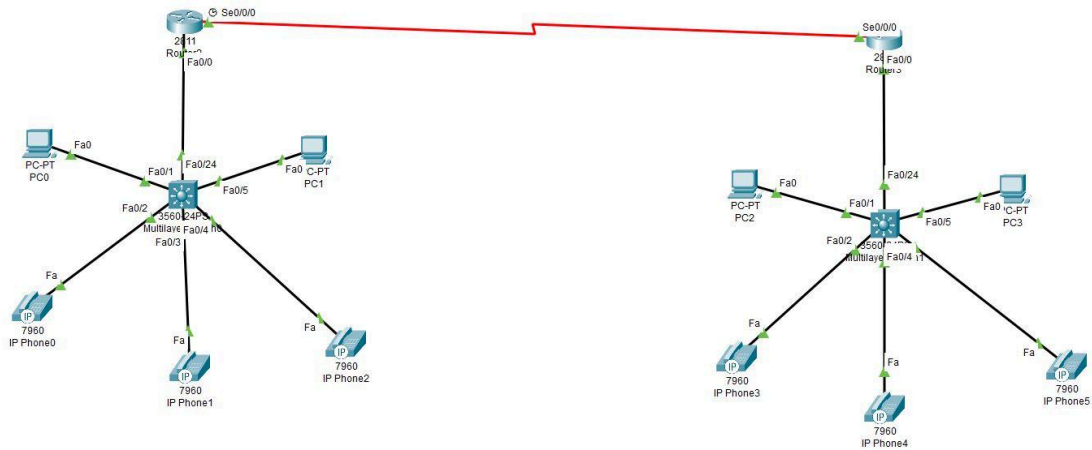


This illustration depicts a VoIP network architecture. In the foreground, three IP phones are connected to a central switch, which is then connected to two routers. The background features a cloud-based network diagram with a computer, server, and database icon, representing the cloud infrastructure used for VoIP services.

CONFIGURACIÓN DE TELEFONIA IP EN CISCO PACKET TRACER

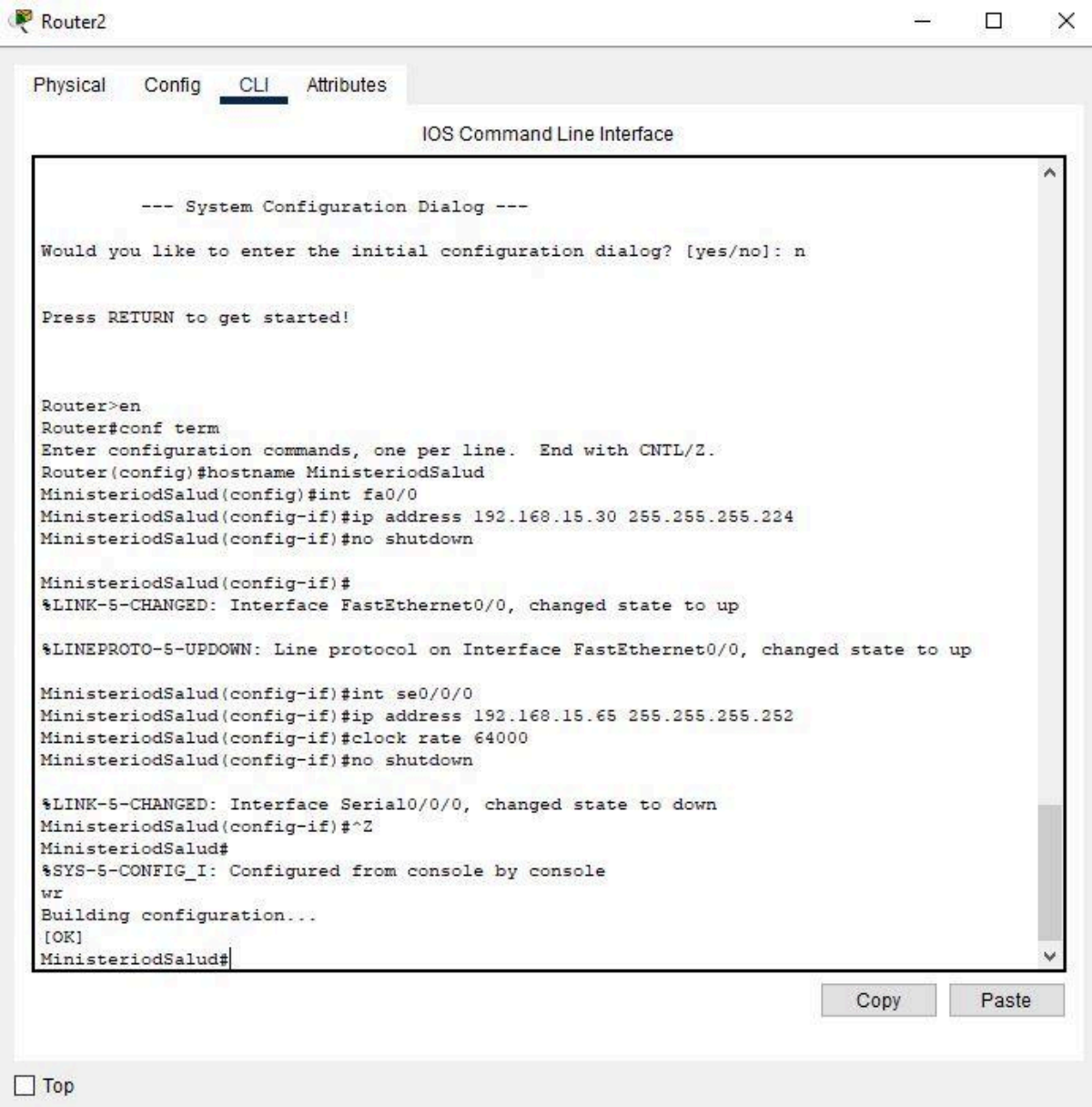
Tenemos la siguiente red en cisco packet tracer, en la que vamos a configurar nuestros routers para poder conectar entre si la telefonia ip de ambas redes.



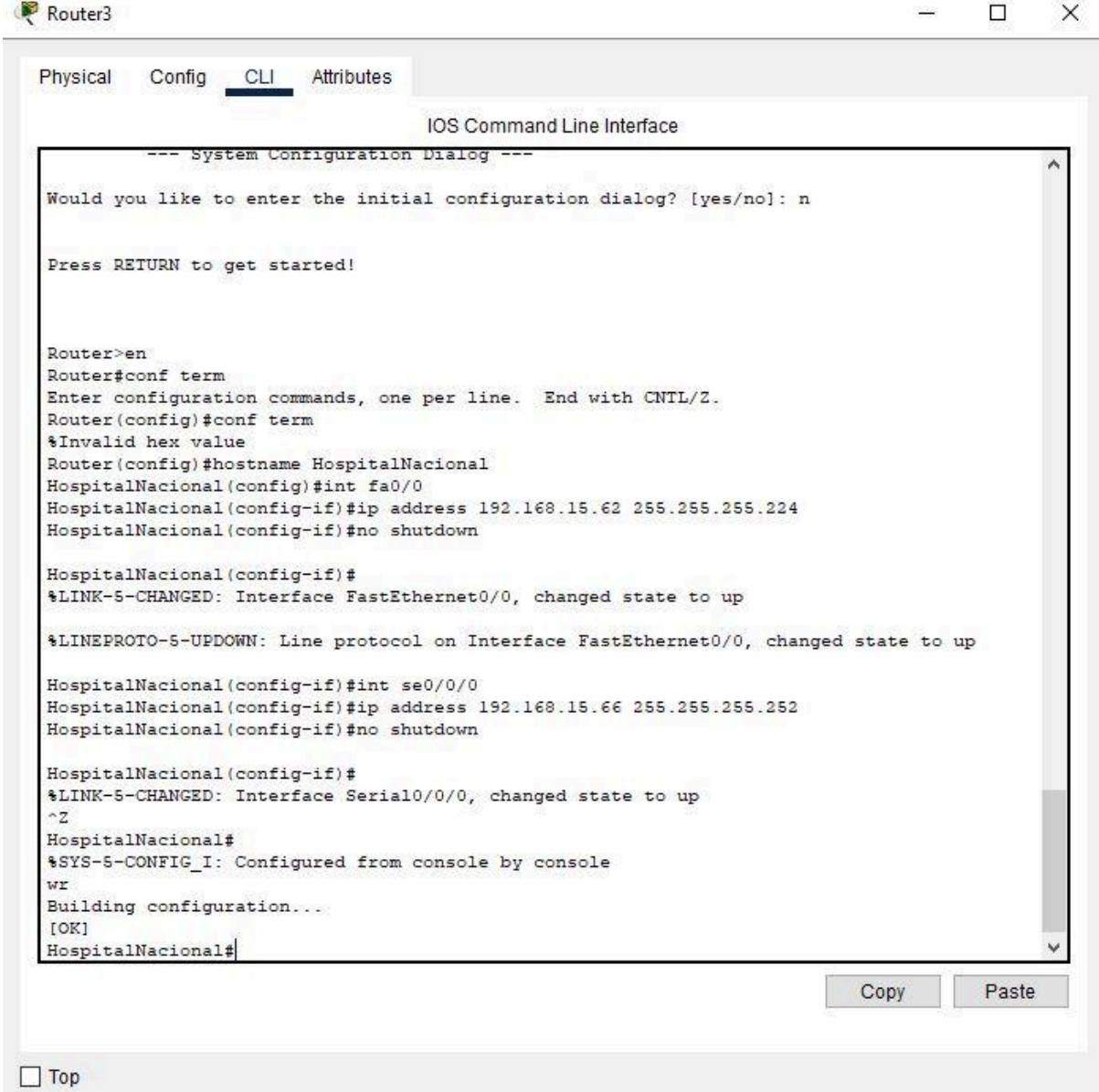
Configuramos interfaces y direcciones ip en los routers

Vamos a tener dos routers 2811, que permiten configuración de telefonía IP. Los nombraremos como Ministerio de Salud y Hospital Nacional respectivamente para poder diferenciarlos. Ambos tendrán 2 interfaces, una FastEthernet con la que se conectarán a la LAN, y una serial con la que estarán conectados entre ellos.

Para el Ministerio de Salud configuraremos la LAN 192.168.15.30/27, y para el serial con el otro Hospital Nacional la 192.168.15.65/30.



Para HospitalNacional configuraremos la LAN 192.168.15.62/27, y para el serial con el otro MinisteriodeSalud la 192.168.15.66/30.



The screenshot shows a Cisco IOS Command Line Interface (CLI) window titled "Router3". The window has tabs for "Physical", "Config", "CLI", and "Attributes", with "CLI" selected. The main area displays the "IOS Command Line Interface" and a "System Configuration Dialog". The dialog asks "Would you like to enter the initial configuration dialog? [yes/no]: n" and "Press RETURN to get started!". The user enters "en" to enter configuration mode. The user then enters "conf term" to enter configuration mode. The user enters "hostname HospitalNacional". The user enters "int fa0/0". The user enters "ip address 192.168.15.62 255.255.255.224". The user enters "no shutdown". The user enters "int se0/0/0". The user enters "ip address 192.168.15.66 255.255.255.252". The user enters "no shutdown". The user enters "Z" to save the configuration. The user enters "wr". The user enters "Building configuration...". The user enters "[OK]". The user enters "HospitalNacional#".

```
Router3
Physical Config CLI Attributes
IOS Command Line Interface
--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]: n
Press RETURN to get started!
Router>en
Router#conf term
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#conf term
%Invalid hex value
Router(config)#hostname HospitalNacional
HospitalNacional(config)#int fa0/0
HospitalNacional(config-if)#ip address 192.168.15.62 255.255.255.224
HospitalNacional(config-if)#no shutdown
HospitalNacional(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
HospitalNacional(config-if)#int se0/0/0
HospitalNacional(config-if)#ip address 192.168.15.66 255.255.255.252
HospitalNacional(config-if)#no shutdown
HospitalNacional(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
^Z
HospitalNacional#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...
[OK]
HospitalNacional#
```

Configuramos DHCP para que los routers asignen IPs a los telefonos y equipos.

Configuraremos el DCHP en ambos routers para evitar errores en la configuración y hacer que esta sea más escalable

Ministerio de Salud

```
MinisteriodSalud#en
MinisteriodSalud#conf term
Enter configuration commands, one per line. End with CNTL/Z.
MinisteriodSalud(config)#ip dhcp pool MinisteriodeSalud
MinisteriodSalud(dhcp-config)#network 192.168.15.0 255.255.255.224
MinisteriodSalud(dhcp-config)#default-router 192.168.15.30
MinisteriodSalud(dhcp-config)#option 150 ip 192.168.15.30
MinisteriodSalud(dhcp-config)#exit
MinisteriodSalud(config)#exit
MinisteriodSalud#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...
[OK]
MinisteriodSalud#
```

Copy

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Hospital Nacional

```
HospitalNacional#conf term
Enter configuration commands, one per line. End with CNTL/Z.
HospitalNacional(config)#ip dhcp pool HospitalNacional
HospitalNacional(dhcp-config)#network 192.168.15.62
% Incomplete command.
HospitalNacional(dhcp-config)#network 192.168.15.62 255.255.255.224
HospitalNacional(dhcp-config)#default-router 192.168.15.62
HospitalNacional(dhcp-config)#option 150 ip 192.168.15.62
HospitalNacional(dhcp-config)#exit
HospitalNacional(config)#exit
HospitalNacional#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...
[OK]
HospitalNacional#
```

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Configuramos el servicio de telefonia Call Manager Express en los routers

Especificaremos el número máximo de telefonos que soportara cada router, y los numeros para marcar

Ministerio de Salud

```
MinisteriodSalud#conf term
Enter configuration commands, one per line. End with CNTL/Z.
MinisteriodSalud(config)#telephony-service
MinisteriodSalud(config-telephony)#max-ephones 5
MinisteriodSalud(config-telephony)#max-dn 5
MinisteriodSalud(config-telephony)#ip source-address 192.168.15.30 port 2000
MinisteriodSalud(config-telephony)#auto assign 4 to 6
MinisteriodSalud(config-telephony)#auto assign 1 to 5
MinisteriodSalud(config-telephony)#exit
MinisteriodSalud(config)#exit
MinisteriodSalud#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...
[OK]
MinisteriodSalud#
```

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Hospital Nacional

```
HospitalNacional#conf term
Enter configuration commands, one per line. End with CNTL/Z.
HospitalNacional(config)#telephony-service
HospitalNacional(config-telephony)#max-ephones 5
HospitalNacional(config-telephony)#max-dn 5
HospitalNacional(config-telephony)#ip source-address 192.168.15.62 port 2000
HospitalNacional(config-telephony)#auto assign 4 to 6
HospitalNacional(config-telephony)#auto assign 1 to 5
HospitalNacional(config-telephony)#exit
HospitalNacional(config)#exit
HospitalNacional#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...
[OK]
HospitalNacional#
```

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Configuramos el switch para permitir el servicio de VOZ

Seleccionamos en cada switch el rango de interfaces en las que queremos habilitar el servicio de voz

```
Switch>en
Switch#conf term
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int range fa0/1-5
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport voice vlan 1
Switch(config-if-range)#wr
^
% Invalid input detected at '^' marker.

Switch(config-if-range)#^Z
Switch#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...
[OK]
Switch#
```

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```

Switch>en
Switch#conf term
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int range fa0/1-5
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport voice vlan 1
Switch(config-if-range)#^Z
Switch#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...
[OK]
Switch#

```

Copy

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Asignamos numeros a los telefonos IP

Ahora le asignamos un numero de marcación a cada telefono. En la red MinisteriodeSalud su extensión sera la 3300x, y en HospitalNacional la 7700x

Ministerio de Salud

IOS Command Line Interface

```

MinisteriodSalud>en
MinisteriodSalud#conf term
Enter configuration commands, one per line. End with CNTL/Z.
MinisteriodSalud(config)#ephone-dn 1
MinisteriodSalud(config-ephone-dn)#%LINK-3-UPDOWN: Interface ephone_dsp DN 1.1, changed
state to up
number 33001
MinisteriodSalud(config-ephone-dn)#number 33001
MinisteriodSalud(config-ephone-dn)#
%IPPHONE-6-REGISTER: ephone-3 IP:192.168.15.2 Socket:2 DeviceType:Phone has registered.

MinisteriodSalud(config-ephone-dn)#exit
MinisteriodSalud(config)#ephone-dn 2
MinisteriodSalud(config-ephone-dn)#%LINK-3-UPDOWN: Interface ephone_dsp DN 2.1, changed
state to up

MinisteriodSalud(config-ephone-dn)#number 33002
MinisteriodSalud(config-ephone-dn)#
%IPPHONE-6-REGISTER: ephone-1 IP:192.168.15.1 Socket:2 DeviceType:Phone has registered.

MinisteriodSalud(config-ephone-dn)#exit
MinisteriodSalud(config)#ephone-dn 3
MinisteriodSalud(config-ephone-dn)#%LINK-3-UPDOWN: Interface ephone_dsp DN 3.1, changed
state to up

MinisteriodSalud(config-ephone-dn)#number 33003
MinisteriodSalud(config-ephone-dn)#exit
MinisteriodSalud(config)#
%IPPHONE-6-REGISTER: ephone-2 IP:192.168.15.3 Socket:2 DeviceType:Phone has registered.
exit
MinisteriodSalud#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...
[OK]
MinisteriodSalud#

```

Hospital Nacional

IOS Command Line Interface

```
HospitalNacional>en
HospitalNacional#conf term
Enter configuration commands, one per line. End with CNTL/Z.
HospitalNacional(config)#ephone-dn 1
HospitalNacional(config-ephone-dn)%%LINK-3-UPDOWN: Interface ephone_dsp DN 1.1, changed
state to up

HospitalNacional(config-ephone-dn)#number 77001
HospitalNacional(config-ephone-dn)#exit
HospitalNacional(config)#ephone-dn
%IPPHONE-6-REGISTER: ephone-3 IP:192.168.15.33 Socket:2 DeviceType:Phone has registered.

% Incomplete command.
HospitalNacional(config)#ephone-dn 2
HospitalNacional(config-ephone-dn)%%LINK-3-UPDOWN: Interface ephone_dsp DN 2.1, changed
state to up

HospitalNacional(config-ephone-dn)#number 77002
HospitalNacional(config-ephone-dn)#exit
HospitalNacional(config)#ephone-dn 2
HospitalNacional(config-ephone-dn)#n
%IPPHONE-6-REGISTER: ephone-1 IP:192.168.15.35 Socket:2 DeviceType:Phone has registered.
HospitalNacional(config-ephone-dn)#exit
HospitalNacional(config)#ephone-dn 3
HospitalNacional(config-ephone-dn)%%LINK-3-UPDOWN: Interface ephone_dsp DN 3.1, changed
state to up

HospitalNacional(config-ephone-dn)#number 77003
HospitalNacional(config-ephone-dn)#
%IPPHONE-6-REGISTER: ephone-2 IP:192.168.15.34 Socket:2 DeviceType:Phone has registered.

HospitalNacional(config-ephone-dn)#exit
HospitalNacional(config)#exit
HospitalNacional#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...
[OK]
HospitalNacional#
```

Configuramos el Dial Peer Mapping

Con el Dial Peer Mapping podemos establecer la conexión entre redes, estableciendo a que red tienen que dirigirse los paquetes en funcion de la extensión que marquemos.

Ministerio de Salud

```
MinisteriodSalud#conf term
Enter configuration commands, one per line. End with CNTL/Z.
MinisteriodSalud(config)#dial-peer voice 1 voip
MinisteriodSalud(config-dial-peer)#destination-pattern 7700.
MinisteriodSalud(config-dial-peer)#session target ipv4:192.168.15.66
MinisteriodSalud(config-dial-peer)#exit+
^
% Invalid input detected at '^' marker.

MinisteriodSalud(config-dial-peer)#exit
MinisteriodSalud(config)#exit
MinisteriodSalud#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...
[OK]
MinisteriodSalud#
```


Hospital Nacional

```
HospitalNacional#conf term
Enter configuration commands, one per line. End with CNTL/Z.
HospitalNacional(config)#dial-peer voice 1 voip
HospitalNacional(config-dial-peer)#destination-pattern 3300
HospitalNacional(config-dial-peer)#destination-pattern 3300.
HospitalNacional(config-dial-peer)#session target ipv4:192.168.15.65
HospitalNacional(config-dial-peer)#exit
HospitalNacional(config)#exit
HospitalNacional#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...
[OK]
HospitalNacional#
```

Establecemos enrutamiento estatico

Finalmente establecemos enrutamiento estatico para permitir que se puedan conectar los equipos de ambas redes

```
MinisteriodSalud#conf term
Enter configuration commands, one per line. End with CNTL/Z.
MinisteriodSalud(config)#ip route 192.168.15.32 255.255.255.224 192.168.15.66
MinisteriodSalud(config)#exit
MinisteriodSalud#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...
[OK]
MinisteriodSalud#
```

```
HospitalNacional>en
HospitalNacional#conf term
Enter configuration commands, one per line. End with CNTL/Z.
HospitalNacional(config)#ip route 192.168.15.0 255.255.255.224 192.168.15.65
HospitalNacional(config)#exit
HospitalNacional#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...
[OK]
HospitalNacional#
```

Prueba de conexión

Una vez finalizada la configuración de nuestra red solo queda comprobar que nuestro servicio de telefonía funciona correctamente. Desde el telefono de Hospital Nacional con numero 77002, marcamos al telefono 33001 que se encuentra en Ministerio de Salud, y vemos que el 33001 recibe la llamada entrante

